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The Modak–Walawalkar Framework for Batteries

Bayesian Cybersecurity

Think Manifolds, Not PDEs. Physics written as Bayesian priors, cut down to a learned manifold, read on a CPU from CAN data alone. LFP and NMC.

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The idea in brief

• **The usual route.** To forecast how a battery ages, write the electrochemistry, derive the equations, and solve them for every cell, every time conditions change. That is slow for a fleet and it still needs parameters nobody has for most packs.

• **The MW route.** Write what a healthy cell is *allowed* to do as Bayesian priors. A variational autoencoder (Pyro on PyTorch) learns the thin set of states those priors permit, a **manifold**. A forecast is then a **walk on that manifold**, and the **distance from it** is the risk score. No equation is solved at run time.

• **What runs today.** One engine, two chemistries (LFP and NMC), 100 Ah-class packs, from an ordinary BMS/CAN log by upload or live feed, on a CPU. A 16-cell pack is analysed in about a second and a half on a laptop.

• **Evidence.** On public data the models forecast state of health within **2.4–3.4 pp median RMSE on 95 LFP cells** and **2.4 pp on NMC** (held-out, 10 of 11 cells in domain), against 12–21 pp for the physics formula alone.

• **Boundary.** Uncertainty bands are still too narrow, cold-temperature NMC is not yet accurate, and no field NMC pack has been tested. Section 7 lists these in full.

How the MW Framework redefines the SPM / SPMe / DFN approach

Dimension	Traditional PyBaMM (SPM / SPMe / DFN)	The MW Framework route
The core math	Solving PDEs: integrates physical mass transport and potential conservation equations across space and time.	Manifold geometry: learns a thin, curved Riemannian state surface using a Bayesian VAE (Pyro on PyTorch).
Runtime compute cost	High to moderate: scales heavily with model complexity; needs numerical solvers (like IDAKLU) that can crash if the equations get stiff.	Extremely low: execution is a simple walk along the manifold. A 16-cell pack takes about 1.5 seconds on a standard laptop CPU.
Data requirements	Lab-grade inputs: dozens of highly specific, cell-isolated electrochemical parameters (diffusion coefficients, reaction rates).	Field-ready logs: runs on ordinary, noisy BMS/CAN log telemetry (voltage, current, temperature).
Handling physical anomalies	Fails to solve: deviations from expected chemistry make the numerical integration break or return unphysical values.	Flags as risk: anomaly detection is built in; the physical distance from the manifold is itself the safety / risk score.

1 · The problem

A battery's health is not a single number you can read off a sensor. It is the slow outcome of several mechanisms at once: film growth on the anode (faster when warm, and roughly with the square root of time), stress from cycling, lithium plating when cold or fast-charged, and particle cracking. Physics-based models such as Doyle–Fuller–Newman describe these well, but they need many cell-specific parameters and a numerical solve for each cell. Black-box machine learning needs no parameters, but it knows nothing about physics: it can forecast a cell that heals itself, or fail badly when a pack runs outside the data it was trained on.

Fleets and storage sites need a third option: physics-respecting, fast enough to run over thousands of cells on ordinary servers, and able to start from the data they already have, the CAN log from the BMS. The same option is needed for NMC, which dominates electric vehicles, as well as for LFP, which dominates stationary storage.

2 · The MW idea: think manifolds, not PDEs

A cell cannot be anywhere in the space of possible telemetry values. Its physics confines it to a thin, curved set, like a ball in a bowl that can only sit on the bowl's surface. The PDEs describe motion on that surface. The surface itself is the more basic object, and if it can be learned directly, the equations need not be derived first.

Step 1: write physics as priors. Each physical fact becomes a probability distribution chosen to carry it:

Battery fact	Prior that carries it
State of charge lies between 0 and 1	Beta distribution
Current cannot exceed the cell's limit	Truncated normal, $I \le I_{max}$
Rate constants are positive and span orders of magnitude (activation energy, film growth, stress)	LogNormal
A mechanism is either active or not (lithium plating, particle cracking)	Bernoulli regime switch
Ageing only moves one way	Non-increasing path on a forward-only time axis
Heat follows Arrhenius and Joule laws; charge is conserved (coulomb closure)	Separate physics priors on one manifold

Step 2: cut the hypercube to a manifold. The priors concentrate probability on the thin set of states a real cell can occupy. A Bayesian VAE written in Pyro learns that set. Its decoder induces a Riemannian metric $G(z) = J^T W J$ on the latent space, so distances and straight lines are measured on the surface, not in the full sixteen-channel space.

Step 3: compute on the manifold. Questions become geometry:

• **Walk.** A forecast is a path along the manifold; no step size, no stiffness, no re-solve. Where the ageing regime switches, the manifold folds and the walk follows the fold: the capacity knee.

• **Distance.** How far a pack's state sits from the manifold measures how physically implausible it is. It needs no examples of past failures.

• **Residuals per prior.** They show which assumption the data rejected, and by how much. A large residual on the monotone-ageing prior while the others stay small points at charge regeneration, as in lithium plating or cell reversal.

The cost of a forecast therefore grows with the dimension of the manifold, not with the size of the space of conceivable telemetry, which is why a pack analysis runs on a CPU in seconds.

3 · The battery engine

Component	What it is
Input	An ordinary 12–24 h BMS/CAN log, converted to a 16-parameter record per cell (voltage, current, temperature, C-rate, depth of discharge, resistance and others). No lab test, no reference cycle.
Model	A Bayesian VAE with physics priors, trained on a synthetic fleet made by a physics generator. A YAML file sets the chemistry (voltage window, resistance, ageing constants) and the fleet's fade rate.
Two forms	**Monotone** (V5B): state of health can only fall or stay flat. **Non-monotone** (V5.1): the same model without that constraint. The engine runs both and routes by domain and state of health.
Engines	**Masked** hides the state-of-health inputs, so the model must infer fade from operating conditions (the headline, field-like test). **Walk** holds the measured value. **Server** walks to an end state set by the physics formula.
Domain check	A cell is in domain when every live input is within 3 standard deviations of training. Outside it, the monotone prior keeps the forecast physical.
Outputs	Per-cell resistance; lithium-inventory loss from the dQ/dV peak shift; a life forecast with a credible window; probability of end of life before N cycles; lever ranking against posterior spread; ten physics integrity laws (MP1–MP10).
Why a short log is enough	This is Kennedy–O'Hagan-style calibration with physical priors in place of Gaussian processes: the family of ageing paths comes from physics and a synthetic fleet, and the log locates the cell on it.

4 · Evidence on open data

All scores are median RMSE of the state-of-health path in percentage points (pp), on cells that never touched calibration or training. A positive bias means optimistic. The task: from one measurement at 100 equivalent full cycles, forecast the full path.

LFP (Sandia and HUST; A123 18650 cells; Preger et al. 2020, Ma et al. 2022):

	Sandia, 18 cells	HUST, 77 cells
Conditions	15, 25, 35 °C; steady fade	30 °C; fast fade with a knee; every cell run past 80 %
Held out	10	40
MW, calibrated	within the 2.4–3.4 pp range once calibrated; outside the training data monotone 3.0 pp vs 20.4 pp for the unconstrained model	2.44 pp monotone, 3.38 pp non-monotone (masked)

	Sandia, 18 cells	HUST, 77 cells
Physics formula alone	about 18–21 pp	18.20 pp
Uncalibrated monotone	—	9.84 pp, 7.7 pp optimistic; calibration to the fleet fixes it

NMC (Sandia NMC/graphite 18650; 21 cells, 10 for calibration, 11 held out):

Model, masked engine	Median RMSE	Notes
NMC-trained monotone, lab scale	**2.41 pp**	10 of 11 cells in domain; all six 25 °C cells within 2.6 pp
NMC-trained monotone, app scale (shipped)	2.51 pp (all 11); 1.09 pp (3 in-domain)	100 Ah-equivalent units; 3 of 11 in domain because cell voltages sit below the generator's NMC window
NMC-trained non-monotone, lab scale	4.15 pp	Monotone is again the sharper tool
LFP production model on the same NMC cells	7.30 pp (6.4 pp optimistic)	A chemistry-specific model is not optional
Physics formula alone	11.67 pp	

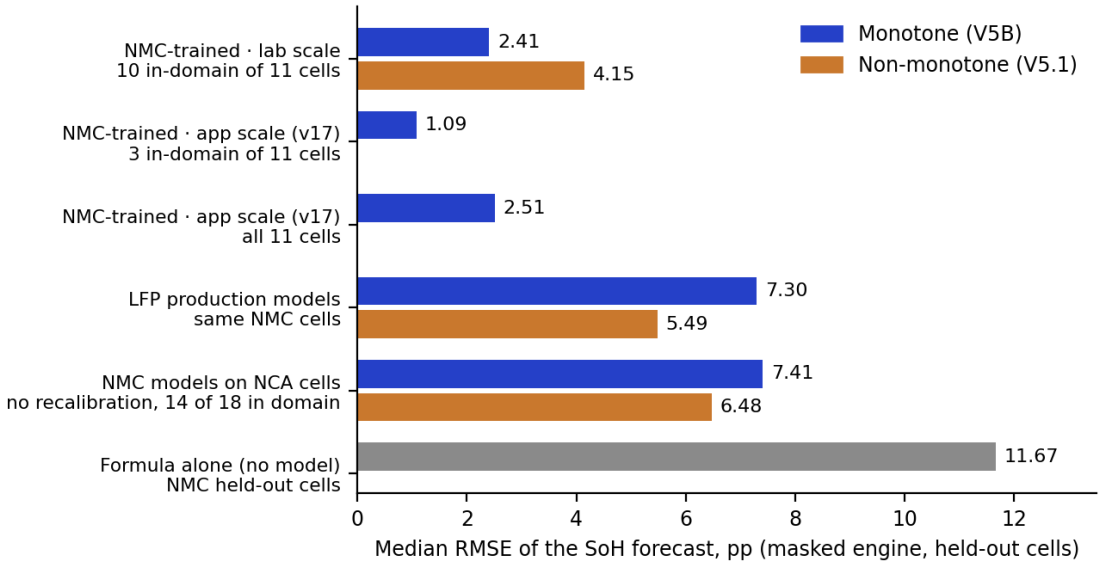


Figure 1. Median RMSE on the held-out NMC cells, masked engine. Lab-scale and app-scale rows use the same 11 cells; the LFP rows are the shipped production checkpoints; the NCA row scores the NMC models on Sandia NCA cells with no recalibration.

What this shows. The same engine, retrained on each chemistry's physics and calibrated on a few cells, forecasts held-out cells of both chemistries to about 2–3 pp. The monotone prior holds up when inputs are unfamiliar. Calibration does not carry across cathodes: the NMC models score 6.5–7.4 pp on NCA cells, no better than the formula. Each new cell family needs its own calibration pass, which is a repeatable step.

5 · Running it: CAN only, on a CPU

• **Inference** runs on a CPU. A 16-cell pack log analysed in 1.5 s on a laptop in the v17 application (Detailed mode, NMC; all forecast, ops, laboratory and AI-summary endpoints returned results with no errors).

• **Training** takes 2–3 hours on one machine for LFP, per the authors; a laptop GPU is a convenience, and LFP training has been run on CPU. NMC models were trained on the same laptop GPU.

• **Two ways in:** upload a CAN CSV, or connect a live CSV feed. The live feed has been tested against a simulated foxBMS-style source. A chemistry switch (LFP or NMC) routes each upload to its own checkpoint, and the application flags a mismatch when a file does not match the model.

• **Scale:** the LFP model is trained for 100 Ah batteries; other capacities are served by request-trained models. The NMC model is trained at 100 Ah-equivalent scale.

6 · Security: physics is hard to forge

Industrial buses such as CAN have no authentication, but physics cannot be faked cheaply. A spoofed state of charge must still match the current that flowed; a faked temperature must still follow Joule heating; manipulated voltages must still relax correctly. A command can be protocol-valid yet physically impossible, or a protection function can be quietly disabled. Both appear as a departure from the manifold: the integrity laws fire, or the MW distance rises above 3 standard deviations. A battery life forecast is not a safety system and cannot see thermal runaway.

7 · Boundaries

Limit	Where it stands
Uncertainty bands	Too narrow on open data: the 90 % band covered 12–23 % of measured points on LFP and about 15 % on NMC. Recalibration is under way. Do not present the band as a probability yet.
End-of-life date	Often missed: monotone reached 80 % on 5 of 18 HUST cells; on NMC the forecast reached it on 1 of 3 cells that did. Non-monotone through the knee is the better route for EOL.
Cold and hot NMC	Both 15 °C cells missed by 9–13 pp and the two 35 °C 1C cells by 4–5 pp; the generator's cold-plating term is small and has no C-rate factor. Not validated.
One NMC family	Only Sandia NMC cells; calibration and test cells share a study. Not a blind test on a different NMC cell. NCA needs recalibration.
Lab cells, not packs	No cell imbalance, BMS noise or CAN step in the open-data scores. No field NMC pack log has been tested.
Physics checks	Thresholds for the ten checks are not yet retuned for NMC.
Not a safety system	A forecast cannot see thermal runaway.

8 · What is new, and what is not

MW is built from established mathematics: Riemannian geometry, variational autoencoders and Bayesian priors. Each ingredient has a named lineage: physics-informed learning, Kennedy–O'Hagan calibration, latent-space reduced-order models, the latent-space curvature of Arvanitidis et al., and distance-to-manifold anomaly scoring. What we have not found in a prior-art search (29 October 2026; companion document) is the combination: physics written deliberately as priors, a learned Riemannian manifold in place of derived equations, forecasts as walks on it, and one distance serving as both the physical model and the risk measure, delivered for batteries from CAN data on a CPU. A web search can find prior art but not prove its absence, and a formal search is a stated next step.

A forward-only time axis gives one more property. Health cannot recover, so a forecast in this model cannot take a cell from 70 % back to 95 %, as a learned transition function might. A path that changes faster than physics allows lies outside the model's causal cone and is inadmissible.

9 · Next steps

1. A second open NMC dataset and a cold-temperature term in the generator. 2. Calibrate the generator's NMC voltage window so more app-scale cells fall in domain. 3. Test an NMC pack log from a real BMS, including the live feed. 4. Retune the ten physics checks for NMC. 5. Widen the uncertainty bands until they cover what they claim. 6. A blind prospective study: life intervals filed before cells reach end of life, scored for coverage. 7. A knee-aware term so the end-of-life date is reached on time.

Sources

Y. Preger et al., “Degradation of commercial lithium-ion cells as a function of chemistry and cycling conditions”, *J. Electrochem. Soc.* 167 (2020) 120532. G. Ma et al., *Energy Environ. Sci.* 15 (2022). BatteryLife processed dataset, Zenodo record 15531867. M. C. Kennedy and A. O'Hagan, *J. R. Stat. Soc. B* 63 (2001). G. Arvanitidis, L. K. Hansen, S. Hauberg, “Latent space oddity”, ICLR 2018. E. Bingham et al., “Pyro”, *JMLR* 20 (2019). Bayesian Cybersecurity, MW Battery Engine validation reports (Sandia + HUST monotone verdict; NMC on Sandia), October 2026.

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